ENGR 225 Homework 4 Questions

Assoc. Prof. Clayton Byers

Assigned: 2 Oct 2026

Due: 8 Oct 2026 (11:59pm on Moodle)

Important notes before you begin:

- Note that this is due the **Thursday night** before the exam!
- Problem 3 will likely require you to solve an equation numerically with Matlab, Mathematica, your fanciest calculator, wolframalpha.com, etc. The point is to get your equations appropriately set up then use an advanced tool to provide the solution.

Problem 1

Determine the components of the support reactions at the fixed support A on the cantilevered beam. (**Ans:** $A_x = -3.46\,kN$ (pointing left), $A_y = 8\,kN$, and $M_A = 20.2\,kN \cdot m$)

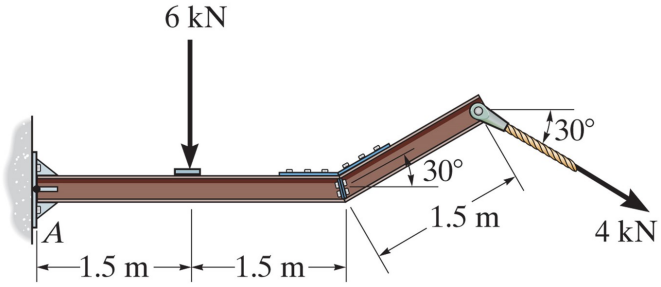


Figure for problem 1.

Problem 2

If the roller at A and the pin at B can support a load up to $4\,kN$ and $8\,kN$, respectively, determine the maximum intensity of the distributed load w, measured in kN/m, such that failure of the supports does not occur. (**Ans:** $w = 2.67\,kN/m$)

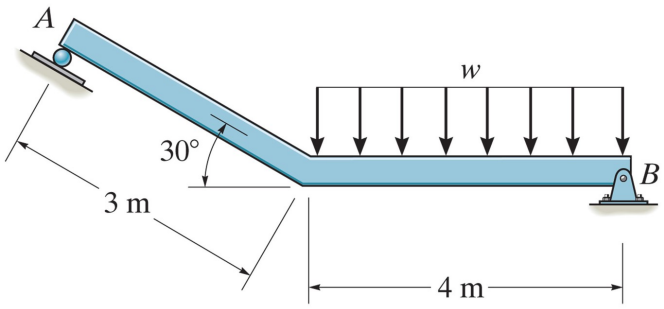


Figure for problem 2.

Problem 3

The $10\,kg$ uniform rod is pinned at end A. If it is also subjected to a couple moment of $50\,N \cdot m$, determine the smallest angle θ for equilibrium. The spring is unstretched when $\theta = 0°$, is free to roll vertically (so it stays horizontally aligned), and has a stiffness of $k = 60\,N/m$. (**Ans:** $\theta = 24.6°$)

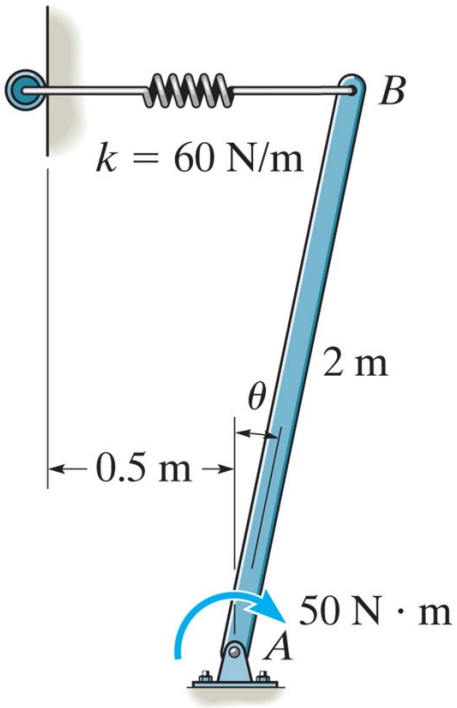


Figure for problem 3.

Problem 4

If $d = 1\,m$ and $\theta = 30°$, determine the normal reaction at the smooth supports and the required distance a for the placement of the roller if $P = 600\,N$. Neglect the weight of the bar. (**Ans:** $N_w = 346\,N$, $N_r = 693\,N$, and $a = 0.650\,m$)

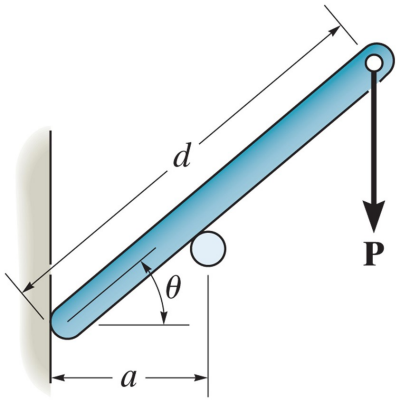


Figure for problem 4.

Problem 5

Determine the components of reaction at the fixed support A. The $400\,N$, $500\,N$, and $600\,N$ forces are parallel to the x, y, and z axes, respectively. (**Ans:** $A_x = 400\,N$, $A_y = -500\,N$, $A_z = 600\,N$, $M_x = 1.23\,kN \cdot m$, $M_y = 750\,N \cdot m$, and $M_z = 0$)

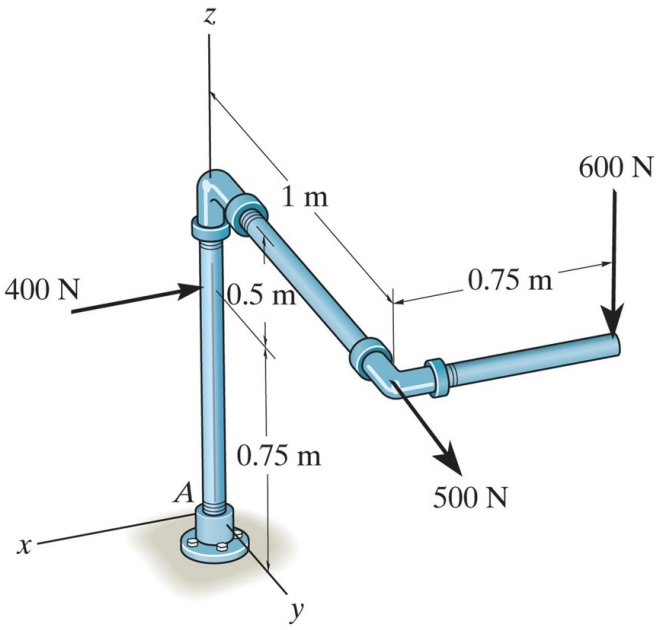


Figure for problem 5.

Problem 6

The uniform concrete slab has a mass of $2400\,kg$ and is suspended by three cables. Determine the tension in each of the three parallel supporting cables when the slab is held in equilibrium in the horizontal plane as shown below. (**Ans:** $T_A = 7.31\,kN$, $T_B = 16.5\,kN$, and $T_C = 14.8\,kN$)

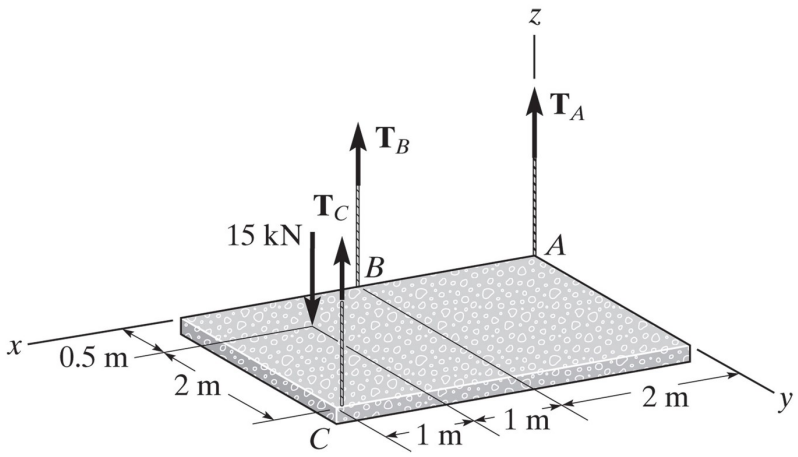


Figure for problem 6.